International Institute of Information Technology, Hyderabad
(Deemed to be University)

Data Systems Exam
Question Paper

90 Minutes :: (90+10) Marks

26th February 2026, 4:30PM

Notes

a) No clarifications will be provided during the exam. Make suitable and justifiable assumptions if needed, and state them.

b) Start the answer to each question on a separate page.

c) **Allowed to bring one handwritten A4 help sheet.**

d) **No calculators or electronic devices**

e) If you do not know the answer for any question or its part question, just state "Do not know" (and nothing else) and you will be given five marks.

f) Note that for this exam, block, disc block, data block, bucket they all mean a disk block that stores some data.

g) Justify all your answers with a few statements of reasoning about the answer, and why it is correct, justification will carry at least 25% of marks.

Questions

1. [20 Marks] Consider the following set of rows with key K values {129, 513, 1025, 2049, 4097, 8193, 16385, 32769, 65537}
   build a linear hash for these key values,
   - assume each block contains exactly one row.
   - you start with two blocks (M=2).

Answer the following.

To retrieve a row with a given key (K=x) where the value x is taken from the above set of key values –

a. **with the linear hash**, what is the number of blocks accessed in the best case, the worst case and the average case.

b. **With the file as unordered heap file**, what is the number of blocks accessed in the best case, the worst case and the average case.

c. **With the file ordered on key**, what is the number of blocks accessed in the best case, the worst case and the average case.

2. [25 Marks] Consider R(B_1, B_2, …, B_n) a relation with **n binary valued attributes (each B_i taking values 0 or 1)**. The attributes are **ordered as** B_1, B_2, …, B_n.

All queries are of the type below **always starting** with B_1, then B_2, and so on till ending with some B_k where **k = 4 to n**. Assume that the result of any such query is at least m rows spanning multiple disk blocks.

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Select * from R
where B1 = v1 and B2 = v2 and … and Bk =vk
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Design an index (do not give out-of-syllabus bitmap index as an answer) to execute these queries efficiently.

3. [10 Marks] Consider (R $JN_{R.A=S.A}$ S), both R and S are sorted on attribute A, A has many duplicate values spanning multiple disk blocks in R and S, how will the sort-merge join algorithm work in this case. Illustrate your answer with an example.

4. [10 Marks] Consider (R $JN_{R.A=S.A}$ S), A is not a key or a foreign key. Let R have secondary index on A, and S is hashed on A.

What will be an efficient algorithm for R join S? Illustrate your answer with an example to show how your algorithm works.

5. [25 Marks] Consider a PowerB-tree with following change -

- at the root node the block size is $2^0*B = B$,
- at first level the block size is $2^1*B = 2*B$,
- at second level the block size is $2^2*B = 4*B$, and so on,
- at i^{th} level the block size is 2^i*B, and so on,
- at last level h the block size is 2^h*B.

The block size doubles for each next level. Assume all blocks for both B tree and PowerB-tree are completely full (not 2/3rd full).

How will the value of p (number of entries in the block, given l_k (length of key attribute), l_p (length of pointer same for both block and record pointer)) change as the number of levels increase. Derive the formula for p.

Given a complete PowerB-tree of level h=3, how many rows will it point to.

How many levels does a B-tree with a block size B require to store the same number of rows.

6. [10 Marks Bonus] How many total marks out of 90 will you get for first five questions? (a) <=30, or (b) 30< total marks <=60, or (c) >60.

****** No more Questions**thankfully! ********

Comprehend the question,
think for a few minutes and more,
start with first principles, if needed
and then answer.

Best of Luck!